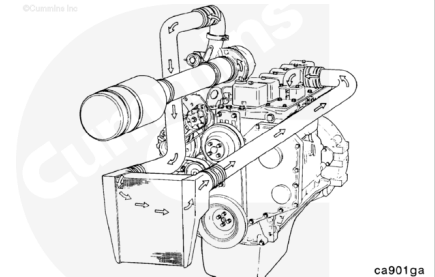


Trainings ▾

General Information

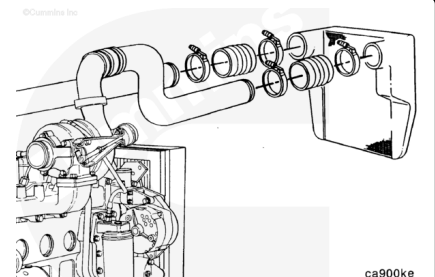
Note : The long-term integrity of the charge air cooler system is the responsibility of the vehicle and component manufacturers; however, the following can be checked by any Cummins Authorized Repair Location.



Clean and Inspect for Reuse

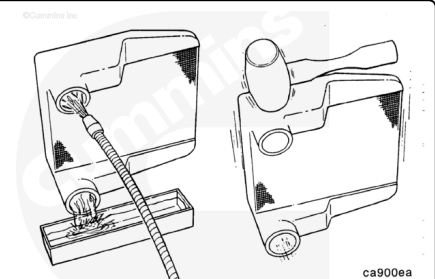
If the engine experiences a turbocharger failure or any other occasion where oil or debris is put into the charge air cooler, the charge air cooler **must** be cleaned and inspected for reuse.

Remove the charge air cooler from the vehicle. Refer to the vehicle manufacturer's instructions.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

⚠ CAUTION ⚠

If internal debris can not be removed, scrap the charge air cooler.

⚠ CAUTION ⚠

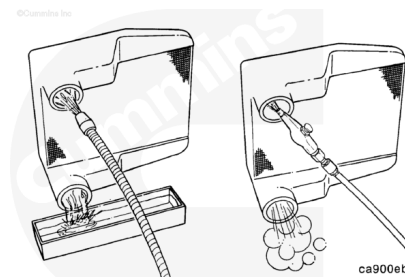
**Do not use caustic cleaners to clean the charge air cooler.
Damage to the charge air cooler can result.**

Flush the charge air cooler internally with solvent in the opposite direction of normal airflow. Shake the charge air cooler, and lightly tap on the end tanks with a rubber mallet to dislodge trapped debris. Continue flushing until all debris or oil is removed.

Use a flashlight and mirror to inspect the charge air cooler for internal debris.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



⚠ CAUTION ⚠

The charge air cooler must be rinsed and dried, and free of solvent, oil, and debris or engine damage will result.

After the charge air cooler has been thoroughly cleaned of all oil and debris with solvent, wash the charge air cooler internally with hot, soapy water to remove the remaining solvent. Rinse thoroughly with clean water.

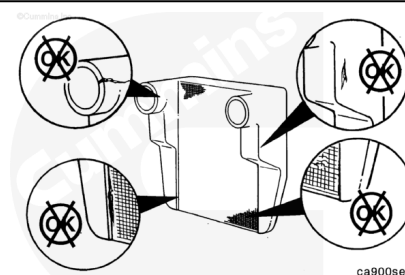
Blow compressed air into the charge air cooler in the opposite direction of normal airflow until the charge air cooler is dry internally.

Inspect the charge air cooler for cracks, holes, or damage.

Inspect the tubes, fins, and welds for tears, breaks, or other damage.

Refer to Procedure 010-024

(</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>) for the leak check procedure.



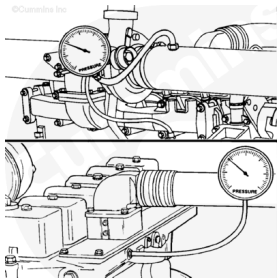
Pressure Test

Install pressure gauge, Part Number ST-1273, to the fitting in the turbocharger outlet.

Install another pressure gauge, Part Number ST-1273, in the intake manifold.

Operate the engine at rated rpm and load. Record the readings on the two gauges.

If the differential pressure is greater than 21 kPa [3 psi], check the charge air cooler for plugging. Clean or replace if necessary.



21 kPa
[3 psi]
Max.

im900sa

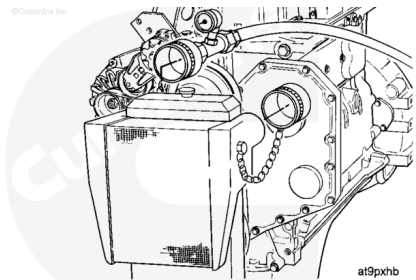
Measurements

	kpa	psi
Differential Pressure:	21	3

Leak Test

⚠ WARNING ⚠

To reduce the possibility of injury from either plug blowing off during the test, secure the safety chains on the test plugs to any convenient capscrew on the radiator assembly. This test must be performed with securely fastened safety chains.



Charge Air Cooler Test Kit, Part Number 3824556

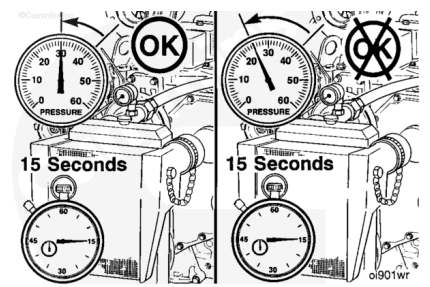
Install a cap over the outlet side of the charge air cooler. Install a pressure gauge, air supply, and air pressure regulator to the inlet side of the cooler.

Apply 207 kPa [30 psi] of air pressure to the charge air cooler. Close the air pressure regulator.

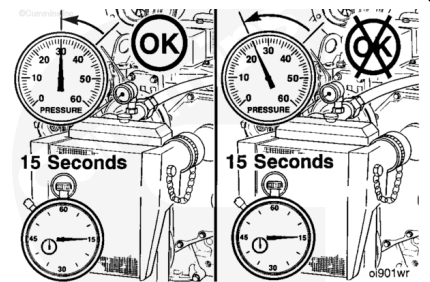
Monitor the pressure gauge and determine the rate of pressure decay with a stopwatch.

If the pressure decay is 49 kPa [7 psi] or less in 15 seconds, the cooler is okay. If the pressure drop is greater than 49 kPa [7 psi] in 15 seconds, check all connections again.

Determine if the pressure decay is caused by a leak in the charge air cooler or from a leaky connection. Use a spray bottle filled with soapy water applied to all hose connections, and watch for bubbles to appear at the location of the leak.



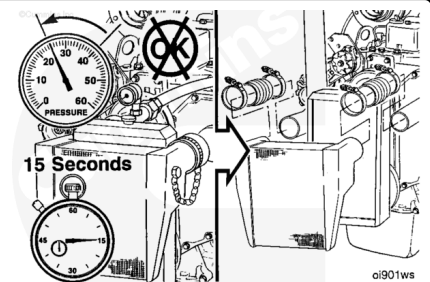
If the pressure decay is caused by a leaky connection, repair the connection and repeat the test. If the leak is within the charge air cooler, repeat the test to verify the accuracy of the pressure decay measurement. Similar pressure decay readings **must** be obtained at least three consecutive tests before the reading can be considered accurate.



If the pressure drop is greater than 49 kPa [7 psi] in 15 seconds, the charge air cooler **must** be replaced.

Refer to the OEM service manual for replacement instructions.

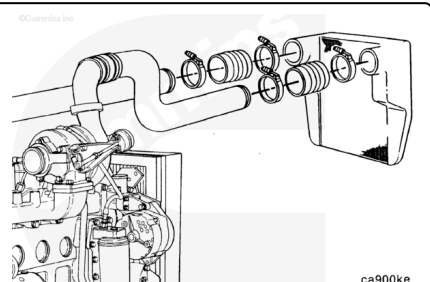
Note : Charge air coolers are **not** designed to be 100-percent leak-free. If the pressure decay is less than 49 kPa [7 psi] in 15 seconds, then the charge air cooler does **not** need to be replaced.



⚠ CAUTION ⚠

Debris trapped in the charge air cooler, if not cleaned, will cause internal engine damage.

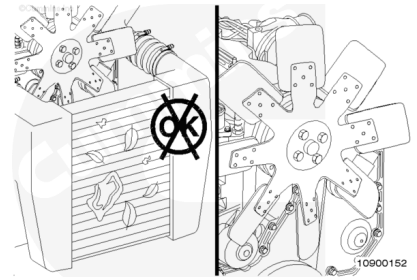
The charge air cooler **must** be cleaned following any turbocharger or air cleaner malfunction.



Temperature Differential Test

Inspect the charge air cooler fins for obstructions to airflow. Remove obstructions such as a winterfront or debris. Manually lock shutters in the open position, if equipped.

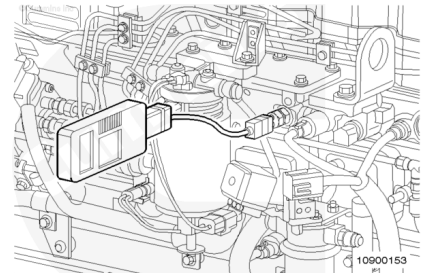
Lock the fan drive in the ON mode to prevent erratic test results. This can be done by installing a jumper wire across the temperature switch.



Install fluke digital thermometer, Part Number 3822666, and thermocouple wire kit, Part Number 3822988, into the intake manifold at the 1/8-inch NPT tap near the air horn connection with the intake manifold.

Another alternative would be to use the monitor mode on the INSITE™ service tool.

Install another thermocouple at the air cleaner inlet to measure ambient air temperature.

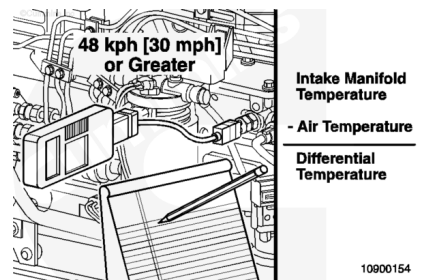


Perform a road test with the engine at peak power and a vehicle speed of 48 kph [30 mph] or greater.

Record the intake manifold temperature and the ambient air temperature.

Calculate the differential temperature:

Intake manifold temperature - ambient air temperature = differential temperature. Maximum differential temperature = 28°C [82°F].



If the temperature differential is greater than the specifications, check the charge air cooler for dirt and debris on the fins and clean as necessary. If the problem still exists, check the charge air cooler for debris in the fins or between the charge air cooler and radiator. Confirm full-fan engagement.

